Example 4.63 By the principle of mathematical induction, prove that, for all integers $n \geq 1$,

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Solution:
Let,

$$P(n) := 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Substituting $n = 1$ in the statement we get, $P(1) = \dfrac{1(1+1)(2(1)+1)}{6} = 1$. Hence, $P(1)$ is true.
Let us assume that the statement is true for $n = k$. Then

$$P(k) = 1^2 + 2^2 + 3^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}.$$

We need to show that $P(k+1)$ is true. Consider

$$\begin{aligned}
P(k+1) &= \underbrace{1^2 + 2^2 + 3^2 + \cdots + k^2} + (k+1)^2 \\
&= P(k) + (k+1)^2 \\
&= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\
&= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\
&= \frac{(k+1)\left(k(2k+1) + 6(k+1)\right)}{6} \\
&= \frac{(k+1)\left(2k^2 + 7k + 6\right)}{6} \\
&= \frac{(k+1)\left[(k+2)(2k+3)\right]}{6} \\
&= \frac{(k+1)\left[((k+1)+1)(2(k+1)+1)\right]}{6}.
\end{aligned}$$

That is,

$$P(k+1) = \frac{(k+1)((k+1)+1)\left(2(k+1)+1\right)}{6}.$$

This implies, $P(k+1)$ is true. The validity of $P(k+1)$ follows from that of $P(k)$. Therefore by the principle of mathematical induction,

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}, \text{ for all } n \geq 1.$$

Example 4.64 Using the Mathematical induction, show that for any natural number n,

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}.$$